

Individual Report Regarding the Project “Energy Efficiency in Railway Scheduling”

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1 My Contribution to the Project

My contribution to the project mostly narrows down to creating the group report and the presentation slides, while Paul was responsible for writing the ASP code to implement the idea. To produce the tables found in the report and the presentation slides I also contributed a Python script to convert the output of Paul’s encodings to a CSV table using regex matching.

2 Challenging Aspects of my Work

As the encoding used a lot of choice rules, and the initial versions used more `#minimize` and `#heuristic` directives, clingo took a lot of time and computational resources to find a solution, which made working on my computer challenging at times, especially on the larger environment with four agents I wanted to use for testing.

Another challenging part was creating a scripts for visualising the tables with GIFs, as debugging took so much time, that I decided to not include a visualisation in the end.

3 Recommendation for future students in this course

Working on a project like this, was more time consuming than I originally expected. For future students I recommend to allocate enough time to work on the upcoming projects. Ideally, they should not have other modules, or their final theses, parallel to this course.